Containers & Kubernetes

Anthos makes multi-cloud easier with new API, support for Azure

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One of the main reasons organizations adopt Anthos is to simplify the management of Kubernetes-based applications across a variety of clouds. And now, with our latest release, we've made multi-cloud management even easier with the general availability of the Anthos Multi-Cloud API.

In addition, in this latest release,

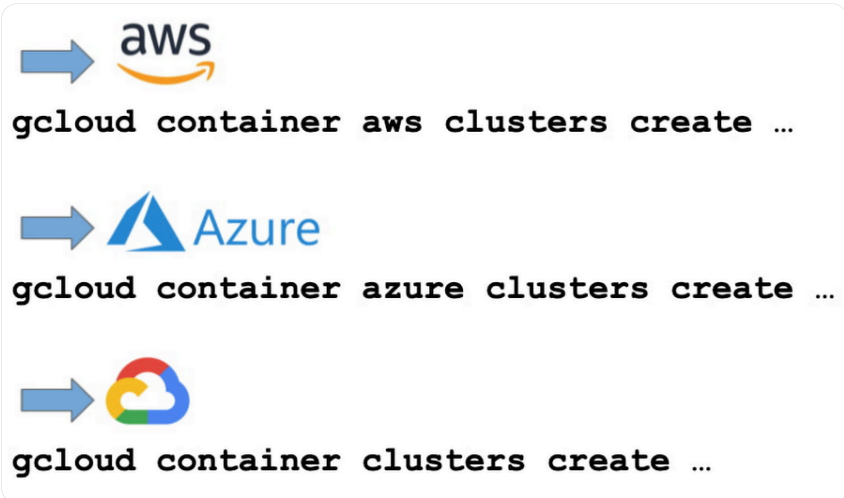
- Support for Anthos clusters[1] running on Azure is also now generally available
- We've added integrated logging and monitoring
- We've introduced [Connect Gateway](https://cloud.google.com) support for unified cluster access, with Terraform and [Kubernetes Config Connector](https://cloud.google.com) support coming soon!

Let's take a deeper look at what you can find in our latest Anthos release.

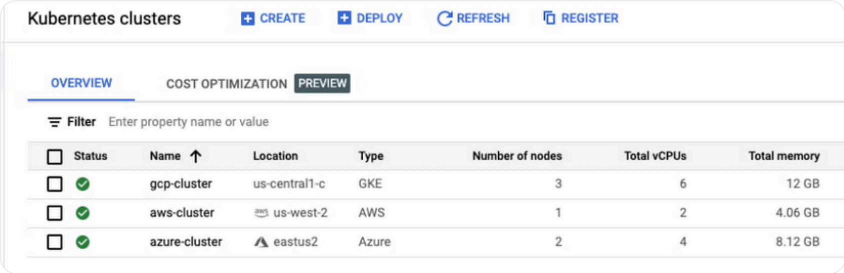
Exploring the Multi-Cloud API

With the latest release of Anthos, we've trimmed our installation footprint and streamlined our cluster management technology to allow you to use a single API for full lifecycle management of Anthos clusters running in AWS or Azure. Compare that to previous releases, which required you to install a management cluster in each cloud. Now, the Anthos Multi-Cloud API, the Google Cloud control plane does all the work! This release standardizes the gcloud CLI for deploying Anthos clusters in AWS, Azure, and GCP (with full Terraform support on the way). Clusters you create in other clouds appear in the Google Cloud Console, creating a centralized management view complete with cluster telemetry and logging.

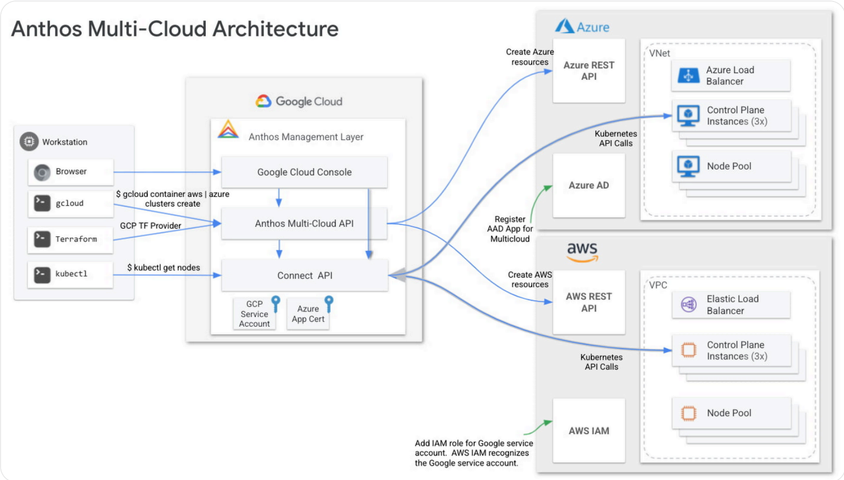
Now, creating a new Anthos cluster on Google Cloud, AWS or Azure is a simple gcloud command:



Here's the associated view from the Cloud Console:



Status	Name	Location	Type	Number of nodes	Total vCPUs	Total memory
✓	gcp-cluster	us-central1-c	GKE	3	6	12 GB
✓	aws-cluster	us-west-2	AWS	1	2	4.06 GB
✓	azure-cluster	eastus2	Azure	2	4	8.12 GB



The Multi-Cloud API performs authentication with each cloud via service account or application registration, and allows clusters to be deployed on existing or newly created VPCs/Vnets. It supports multiple machine types in each cloud, with plans to support even more soon (AWS, Azure). As a reminder, Anthos clusters on Azure or AWS integrate with each respective cloud's native KMS, storage facilities, and load balancing.

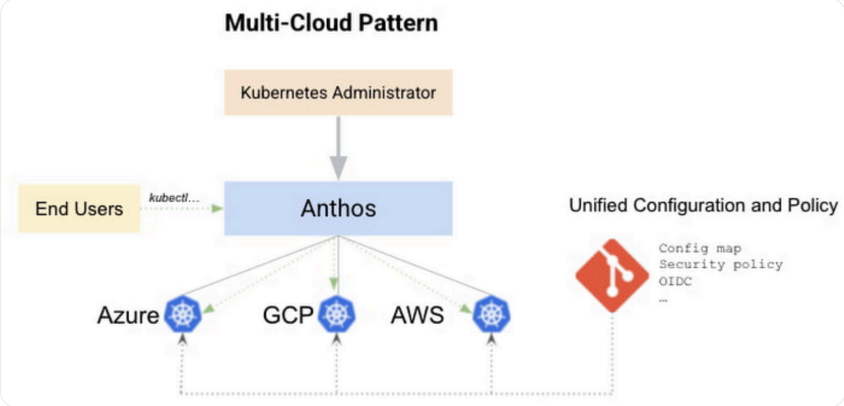
Using Connect Gateway to connect to Anthos clusters in AWS and Azure

Connect gateway allows you to interact with your Anthos clusters securely, and now it works with Anthos clusters running on AWS and Azure too. Cluster commands are routed through a GCP Service to your clusters over an encrypted connection, removing the need for end users to use a VPN.

Putting together a multi-cloud strategy

Operationalizing Google-managed Kubernetes clusters in all three major clouds is now much easier with the release of the Multi-Cloud API. The next step is to apply configuration governance and policy controls to the clusters which will create safe and secure deployment landing zones for your applications regardless of the environment.

For one thing, you can now leverage Anthos Configuration Management (ACM), which automates policy and security at scale for Kubernetes clusters whether they are running on-premises, on GCP, and on other public clouds. ACM synchronizes your clusters to a git repository that contains your business specific configurations and policies. Developers can launch their applications by adding configuration files to the ACM repo or they can use their existing CD tooling. In either case, by using ACM, you can be sure security and governance is applied uniformly across your fleet of clusters.



Meanwhile, Cloud Run for Anthos and Anthos Service Mesh offer tremendous value to organizations looking to optimize and secure Kubernetes-based workloads. Cloud Run for Anthos enables container-based application deployments that scale to zero with predictable costs in your own clusters while making use of existing CI/CD pipelines and security tooling. Anthos Service Mesh brings advanced application networking capabilities to your services and valuable inter-cluster communication telemetry, and is designed to work on Anthos clusters running on GKE, AWS and Azure. These Anthos capabilities are critical to businesses that manage microservice-based applications at scale; look for them to be released in the coming months!

Get started today

Anthos clusters are enterprise-grade Kubernetes clusters that are entirely supported by Google Cloud — and now running them in AWS and Azure is a seamless experience. To get started, check out our Install Anthos Clusters on AWS or Azure guide.

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1. An Anthos cluster refers to a Google-managed Kubernetes cluster that can run outside of Google Cloud.

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